

FORCED HARMONIC MOTION IN UNDAMPED 2ND ORDER LINEAR ODE

- Second order ODEs
- Constant coefficients
- But now, we have a forcing term.

[With the forcing term] m
I'm "pushing" this
motion.

- At a frequency ω .
- Forcing term = $\cos \omega t$

Big Questions:

- Are ω & ω_n close or well-separated?

→ Governs whether the bridge you are walking on oscillates too much & eventually falls

- Are ω & ω_n equal?

[Resonance]
 $\omega = \omega_n$

Recall:

→ We discussed previously,

→ Free harmonic motion.

$$m \frac{d^2 y}{dt^2} + ky = 0$$

[No forcing term]

→ Two null solutions

→ [Natural frequency] $\omega_n = \sqrt{\frac{k}{m}}$

$$m \frac{d^2 y}{dt^2} + ky = \cos \omega t$$

- I'm looking for a particular solution (y_p)
- Two null solutions

① Find a particular solution $[y_p(t)]$:

y_p $\rightarrow$ forced response for input, $\cos(\omega t)$

Let $y_p = A \cos \omega t$ where A is a constant.

Recall:

$$m \frac{d^2 y}{dt^2} + ky = \cos(\omega t)$$

$$\Rightarrow m \frac{d^2 y_p}{dt^2} + k y_p = \cos(\omega t)$$

Substitute $y_p = A \cos(\omega t)$

$$\Rightarrow -m A \omega^2 \cos(\omega t) + k A \cos(\omega t) = \cos(\omega t)$$

$$\Rightarrow (k - m \omega^2) A \cos(\omega t) = \cos(\omega t)$$

$$\Rightarrow (k - m \omega^2) A = 1 \Rightarrow A = \frac{1}{k - m \omega^2}$$

$$\therefore y_p = \left(\frac{1}{k - m \omega^2} \right) \cos(\omega t)$$

② Find the homogenous solution $[y_h(t)]$:

$$m \frac{d^2 y_h}{dt^2} + k y_h = 0$$

$$\Rightarrow y_h = c_1 e^{s_1 t} + c_2 e^{s_2 t}$$

$$\therefore y_h = c_1 e^{i \omega_n t} + c_2 e^{-i \omega_n t}$$

characteristic equation: $m s^2 + k = 0$

$$\Rightarrow s^2 = \pm \sqrt{-\frac{k}{m}} \Rightarrow s_1 = +i \sqrt{\frac{k}{m}} \Rightarrow s_1 = +i \omega_n$$

$$\Rightarrow s_2 = -i \sqrt{\frac{k}{m}} \Rightarrow s_2 = -i \omega_n$$

$$y_h = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}$$

$$= C_1 \cos(\omega_n t) + C_1 i \sin(\omega_n t)$$

$$+ C_2 \cos(\omega_n t) - C_2 i \sin(\omega_n t)$$

$$= (C_1 + C_2) \cos(\omega_n t) + (C_1 - C_2) i \sin(\omega_n t)$$

$$y_h = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

Complete solution: $y(t) = y_p(t) + y_h(t)$

$$y(t) = \left[\frac{\cos(\omega t)}{k - m\omega^2} \right] + C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

$$= \left[\frac{\cos(\omega t)}{m \left(\frac{k}{m} - \omega^2 \right)} \right] + C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

$\because \omega_n^2 = \frac{k}{m}$

$$y(t) = \left[\frac{\cos(\omega t)}{m(\omega_n^2 - \omega^2)} \right] + C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

We get our C_1 & C_2 from initial states $y(0)$ & $y'(0)$

You can see that the denominator has the $m(\omega_n^2 - \omega^2)$ term.

You can start seeing the whole point of resonance or near-resonance behavior.

If ω_n & ω are close to each other, the frequency-squared difference will be small, but the effect is large!

$$\left\{ \begin{array}{l} \text{Frequency} \\ \text{Response} \end{array} \right\} = \frac{1}{m(\omega_n^2 - \omega^2)}$$